California Housing Price Prediction using Linear Regression

Task 1 Report | AI & ML Internship

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1. Project Overview & Objective

In this project, I built a Linear Regression model to predict house prices using the California Housing dataset. The main goal was to practice the end-to-end Machine Learning pipeline: loading data, doing Exploratory Data Analysis (EDA), splitting the dataset, training a simple regression model, evaluating its performance, and visualizing the results.

2. Dataset & Exploratory Data Analysis (EDA)

The dataset used is the built-in California Housing dataset from scikit-learn. It contains information about block groups in California based on the 1990 census data.

- Target Variable: MedHouseVal (Median house value in $100,000s).

- Features: 8 numerical attributes including median income (MedInc), house age (HouseAge), average rooms (AveRooms), and location coordinates (Latitude, Longitude).

- Data Quality Check: I checked for missing values using df.info(), and the dataset was clean with no null entries.

- Key Finding: Looking at the correlation matrix, the feature with the strongest positive relationship to house prices was MedInc (Median Income), which logically makes sense as wealthier areas tend to have higher house values.

3. Model Implementation & Workflow

I implemented the model in Python using pandas, scikit-learn, matplotlib, and seaborn.

1. Train/Test Split: I split the data into 80% for training (X_train, y_train) and 20% for testing (X_test, y_test) with random_state=42 for reproducible results.

2. Model Training: I used LinearRegression() from sklearn.linear_model and fitted it on the training data.

3. Prediction: The model generated price predictions (y_pred) for the unseen test set.

4. Model Evaluation & Results

To evaluate how well the model performed, I calculated three standard regression metrics:

- MAE (Mean Absolute Error): 0.5332 (On average, the model's predictions are off by ~$53,320)

- RMSE (Root Mean Squared Error): 0.7456 (Penalizes larger errors, showing standard deviation of residuals)

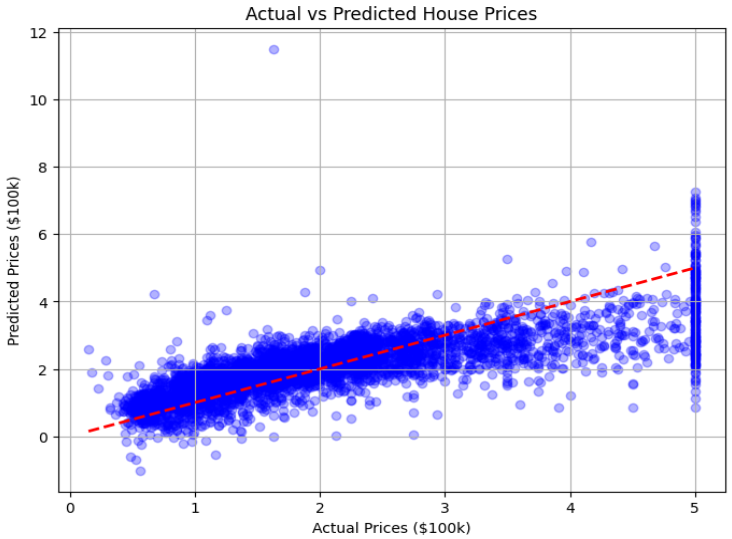
- R^2 Score: 0.5758 (The model explains ~57.6% of the variance in house prices)

I plotted the Actual vs. Predicted values using a scatter plot. Most points clustered around the diagonal line, showing that the model captures the overall trend, though there are some visible outliers, especially near the higher price capped values.

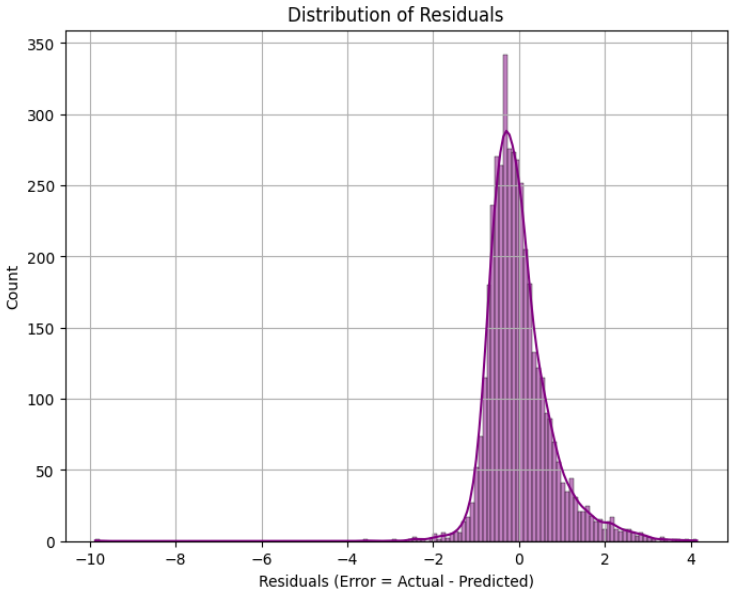
Visualizing Results:

I plotted the Actual vs. Predicted values using a scatter plot. Most points clustered around the diagonal line, showing that the model captures the overall trend, though there are some visible outliers, especially near the higher price capped values.

1.Actual Vs. Predicted:



2.Distribution of Residuals:



5. Conclusion & Future Improvements

The simple Linear Regression model served as a good baseline with an R^2 score of ~0.58. While it captures general patterns well, there is room for improvement.

Ideas for Future Work:

- Try Advanced Algorithms: Test non-linear models like Random Forest Regressor or XGBoost to capture complex feature interactions.

- Feature Engineering: Create new features (e.g., bedrooms per room ratio) or handle geographical coordinates better.

- Outlier Removal: Handle extreme values and price capping to reduce prediction errors.